Homotopy Classification of Vector Bundles

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1 Introduction

Vector bundles arise naturally across geometry and topology as a way of attaching a vector space continuously to each point of a topological space. They provide the precise language for describing tangent spaces on manifolds, physical fields, and families of linear structures parameterized by a space. Understanding when two vector bundles are equivalent and how they differ is therefore a fundamental problem connecting topology, geometry, and algebra.

The classification problem for vector bundles asks the following: given a topological space B, how can we describe all vector bundles over B up to isomorphism? A surprising and powerful answer is that vector bundles can be classified using purely homotopy-theoretic data. In particular, each vector bundle can be obtained as a pullback of a single universal bundle defined over a special classifying space called the Grassmannian. Consequently, the set of isomorphism classes of vector bundles corresponds bijectively to the set of homotopy classes of maps into this space. This transforms the geometric classification problem into a problem in homotopy theory.

The purpose of this project is to develop the theory of vector bundles to rigorously establish this correspondence. We build the theory from the ground up, focusing on the explicit geometric construction of Gauss maps and the topological machinery—such as partitions of unity and paracompactness—required to globalize these constructions.

It is important to note that the classification theory established here is the prerequisite for the theory of characteristic classes. While the broader scope of this project includes the study of these algebraic invariants, this written report focuses exclusively on the first, foundational component: the rigorous proof of the classification theorem. By limiting the scope to the explicit construction of the correspondence $\mathrm{Vect}_k(B) \cong [B, G_k(\mathbb{R}^\infty)]$, we ensure that the geometric and topological foundations are established with full mathematical precision, providing the necessary existence guarantees for any future study of characteristic classes.

The structure of this report is as follows:

In Section 2, we lay the foundations of the theory. We begin by reviewing basic definitions and establishing the crucial notion of bundle isomorphisms. We then introduce the operation of the pullback (or induced bundle), which is the primary mechanism for relating bundles over different spaces. A central part of our discussion focuses on homotopy invariance, the principle that homotopic maps induce isomorphic bundles, which forms the injectivity half of our main theorem. Finally, we construct the infinite Grassmannian and the universal bundle, and use the method of Gauss maps to prove the surjectivity half. This culminates in the proof of the classification theorem.

The main references are [Hus94], [Hat17].

2 Vector Bundles and Classification

To illustrate the idea of a vector bundle, consider the circle S^1 as a base space. To each point $x \in S^1$, we attach a line segment representing a vector space $\mathbb{R}$, called the fiber. If we attach these fibers such that they undergo a half-twist (rotation by π) as one travels a full loop around the circle, the collection of all these fibers forms a continuous surface known as the Möbius strip.

Figure 1 visualizes this construction: the left side shows the individual fibers twisting as they traverse the circle, while the right side shows the resulting total space. This structure—a family of vector spaces parameterized continuously by a base space—is the main idea of a vector bundle.

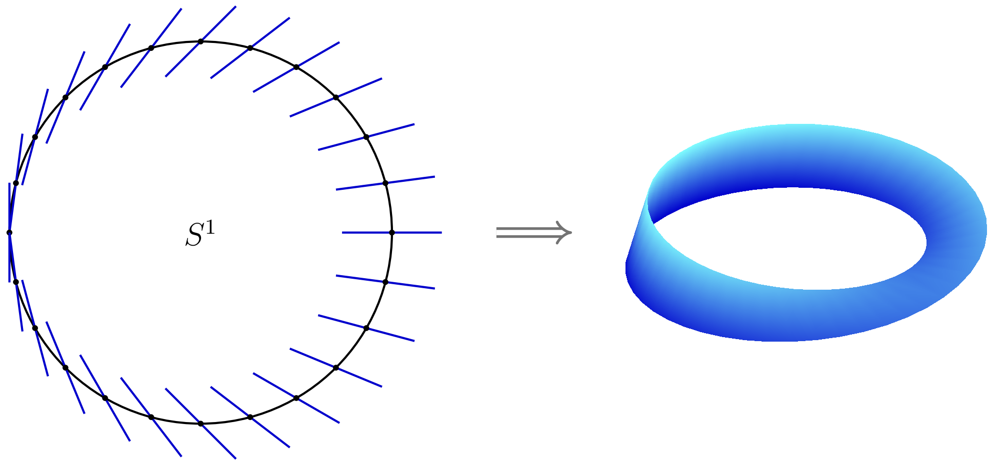

Figure 1: A discrete visualization of fibers twisting over S^1 and the resulting continuous total space, the Möbius strip (generated by Gemini).

Unlike the cylinder, which is the (trivial) bundle $S^1 \times \mathbb{R}$, the Möbius bundle is twisted. Locally, over small arcs of the circle, it looks exactly like a product $U \times \mathbb{R}$, but globally, the twist prevents a decomposition as $S^1 \times \mathbb{R}$.

Associated with this structure is a natural projection map

$$\pi : E \to S^1,$$

sending each point in the total space E, which is the Möbius strip, to the point on the circle to which it is attached. This example illustrates the essential features of a vector bundle: a base space, fibers, a continuous projection, and local triviality, while demonstrating that the total space may be globally distinct from a simple product.

2.1 Basic Notions and Results

Having some idea of what a vector bundle should be, let us now begin to develop the theory. We start with the definition:

Definition 2.1. Let E and B be topological spaces, and let $p : E \to B$ be a continuous[1] map. A real vector bundle of rank[2] k[3] is a triple $\xi = (E, p, B)$, satisfying the following conditions:
(i) For each $b \in B$, the fiber $p^{-1}(b)$ is a k-dimensional vector space over $\mathbb{R}$.
(ii) For each $b \in B$, there exists an open neighborhood $U \subseteq B$ of b and a homeomorphism

$$h : p^{-1}(U) \longrightarrow U \times \mathbb{R}^k$$

such that the diagram

$$\begin{tikzcd} p^{-1}(U) \arrow[rr, "h", "\cong"'] \arrow[dr, "p"'] & & U \times \mathbb{R}^k \arrow[dl, "\mathrm{pr}_1"] \\ & U & \end{tikzcd}$$

commutes and for every $x \in U$, the restriction $h|_{p^{-1}(x)} : p^{-1}(x) \to \{x\} \times \mathbb{R}^k$ is a vector space isomorphism.

The map p is called the projection, the space B the base space, E the total space, and each $p^{-1}(b)$ a fiber. The homeomorphism h is called a local trivialization.

The definition for a complex vector bundle can be given analogously by replacing $\mathbb{R}$ with $\mathbb{C}$.

Remark 2.2. From the definition, we observe that since every vector space contains at least the zero vector, the fibers are non-empty, and thus the projection p is surjective. In the specific case of a rank 0 bundle, each fiber consists of a single point (the zero vector). Consequently, p is a bijection, and one can straightforwardly verify that it is, in fact, a homeomorphism.

[1] From now on, a map $f : X \to Y$ will mean a continuous map unless otherwise stated.
[2] We will use the words rank and dimension interchangeably in the following discussion.
[3] In the following, we will usually not mention the rank of the vector bundle, as the results established will be valid for any finite dimension k unless explicitly stated otherwise.

The following example, though simple, is central to the theory, as local triviality asserts that every vector bundle looks like this one locally:

Example 2.3 (The Trivial Bundle)**.** Let B be a topological space. The triple $\xi = (B \times \mathbb{R}^k, p, B)$, where $p = \mathrm{pr}_1$ is the projection onto the first factor, is a vector bundle of rank k. It admits a single global trivialization (the identity map) and is called the trivial bundle over B.

The next example is familiar to anyone who has studied smooth manifolds:

Example 2.4 (The Tangent Bundle)**.** Let M be a smooth manifold of dimension n. The tangent bundle TM is defined as the disjoint union of tangent spaces:

$$TM = \bigsqcup_{p \in M} T_pM.$$

The projection map $\pi : TM \to M$ sends a tangent vector $v \in T_pM$ to its base point p. Coordinate charts on M induce natural local trivializations $TM|_U \cong U \times \mathbb{R}^n$, giving TM the structure of a rank-n vector bundle. For a detailed construction, see [Lee12].

Before proceeding further, it is useful to consider restrictions of a bundle to a subset of the base space.

Definition 2.5 (Restriction of a Bundle)**.** Let $\xi = (E, p, B)$ be a vector bundle and let $A \subseteq B$ be a subspace of the base space. The restriction of ξ to A, denoted by $\xi|_A$, is the bundle

$$\xi|_A = (p^{-1}(A), p|_{p^{-1}(A)}, A).$$

The total space $p^{-1}(A)$ is endowed with the subspace topology from E. The fibers of $\xi|_A$ are exactly the fibers of ξ over points in A, preserving their vector space structure. It is straightforward to verify that local trivializations of ξ restricted to $A \cap U$ provide local trivializations for $\xi|_A$, ensuring it is indeed a vector bundle.

Remark 2.6. The restriction can be viewed as a special case of the induced bundle introduced in Section 2.5. If $i : A \hookrightarrow B$ is the inclusion map, then $\xi|_A$ is naturally isomorphic to the pullback bundle $i^*(\xi)$.

2.2 The Universal Bundle

In this subsection, we introduce the object that will be the central focus of our discussion: the universal bundle.

Our starting point is the Stiefel manifold $V_k(\mathbb{R}^n)$, defined as the space of all k-tuples of orthonormal vectors in $\mathbb{R}^n$. This can be viewed as a subspace of $(S^{n-1})^k$, the product of k copies of the $(n-1)$-dimensional unit sphere. Consider the map $f : (S^{n-1})^k \to \mathbb{R}^{k^2}$ given by the matrix of inner products:

$$f(v_1, \dots, v_k) = \big((v_i|v_j)\big)_{1\leq i,j\leq k},$$

where $(\cdot|\cdot)$ denotes the standard Euclidean inner product. The Stiefel manifold $V_k(\mathbb{R}^n)$ is precisely the preimage of the identity matrix $I_k \in \mathbb{R}^{k^2}$ (viewing $\mathbb{R}^{k^2}$ as the space of $k \times k$ matrices). Since points are closed sets in the Hausdorff space $\mathbb{R}^{k^2}$ and f is continuous, $V_k(\mathbb{R}^n)$ is a closed subset of $(S^{n-1})^k$. Because the product of spheres is compact, we conclude that $V_k(\mathbb{R}^n)$ is compact.

We now generalize the projective space $\mathbb{R}P^n$ using the Stiefel manifold. Let $G_k(\mathbb{R}^n)$ denote the set of all k-dimensional vector subspaces of $\mathbb{R}^n$, called the Grassmann manifold. There is a natural surjective map $q : V_k(\mathbb{R}^n) \to G_k(\mathbb{R}^n)$ sending an orthonormal k-tuple to the subspace it spans. We topologize $G_k(\mathbb{R}^n)$ with the quotient topology induced by q. An immediate consequence is that $G_k(\mathbb{R}^n)$ is compact, as it is the continuous image of the compact space $V_k(\mathbb{R}^n)$.

Remark 2.7. The Stiefel and Grassmann spaces are indeed manifolds, but this structure will not be required for our classification.

Considering the natural inclusions $\mathbb{R}^n \subseteq \mathbb{R}^{n+1} \subseteq \dots$, we let $\mathbb{R}^\infty = \bigcup_n \mathbb{R}^n$ denote the vector space of infinite sequences of real numbers with only finitely many nonzero terms. We endow this space with the direct limit topology: a subset $U \subseteq \mathbb{R}^\infty$ is open if and only if the intersection $U \cap \mathbb{R}^n$ is open in $\mathbb{R}^n$ for every n.

Observing that the natural inclusions $\mathbb{R}^n \subseteq \mathbb{R}^{n+1} \subseteq \dots$ induce corresponding inclusions $G_k(\mathbb{R}^n) \subseteq G_k(\mathbb{R}^{n+1}) \subseteq \dots$, we define the infinite Grassmannian as the union $G_k(\mathbb{R}^\infty) = \bigcup_n G_k(\mathbb{R}^n)$, representing the set of all k-dimensional subspaces of $\mathbb{R}^\infty$. We similarly topologize this space with the direct limit topology.

We now define the canonical vector bundle γ_k^n over the finite Grassmannian $G_k(\mathbb{R}^n)$. Let

$$E_k(\mathbb{R}^n) = \{(V, v) \in G_k(\mathbb{R}^n) \times \mathbb{R}^n \mid v \in V\}.$$

This space comes equipped with the natural projection map $p : E_k(\mathbb{R}^n) \to G_k(\mathbb{R}^n)$ defined by $p(V, v) = V$. As before, the inclusions $\mathbb{R}^n \subseteq \mathbb{R}^{n+1} \subseteq \dots$ induce inclusions of total spaces, allowing us to define the bundle space $E_k(\mathbb{R}^\infty) = \bigcup_n E_k(\mathbb{R}^n)$, endowed with the direct limit topology.

We denote this limit bundle by $\gamma_k = (E_k(\mathbb{R}^\infty), p, G_k(\mathbb{R}^\infty))$ and call it the universal bundle. The following result confirms that γ_k^n and γ_k are indeed vector bundles.

Proposition 2.8. The projection map $p : E_k(\mathbb{R}^n) \to G_k(\mathbb{R}^n)$ defines a vector bundle of rank k for both finite n and $n = \infty$.

Proof. First suppose $n < \infty$. For any subspace $V \in G_k(\mathbb{R}^n)$, let $\pi_V : \mathbb{R}^n \to V$ be the orthogonal projection onto V. Define the set

$$U_V = \{V' \in G_k(\mathbb{R}^n) \mid \dim(\pi_V(V')) = k\}.$$

Note that $V \in U_V$, so these sets cover the Grassmannian as V varies. We first show that U_V is an open subset of $G_k(\mathbb{R}^n)$.

Recall that $G_k(\mathbb{R}^n)$ carries the quotient topology from the Stiefel manifold $V_k(\mathbb{R}^n)$. The preimage of U_V in $V_k(\mathbb{R}^n)$ consists of orthonormal k-tuples $(v_1, \dots, v_k)$ such that the projected vectors $\pi_V(v_1), \dots, \pi_V(v_k)$ are linearly independent in V. Let us fix a basis for V and let M be the $k \times n$ matrix representing π_V with respect to the standard basis of $\mathbb{R}^n$ and this basis of V. Then the vectors $\pi_V(v_1), \dots, \pi_V(v_k)$ are linearly independent if and only if the $k \times k$ matrix whose columns are $Mv_1, \dots, Mv_k$ has nonzero determinant. This determinant is clearly a continuous function of the coordinates of $v_1, \dots, v_k$. Thus, the preimage of U_V is an open set in $V_k(\mathbb{R}^n)$. By the definition of the quotient topology, it follows that U_V is open in $G_k(\mathbb{R}^n)$.

Next define $h : p^{-1}(U_V) \to U_V \times V \cong U_V \times \mathbb{R}^k$ by $h(V', v) = (V', \pi_V(v))$. Clearly, h is a bijection because the projection restricts to a linear isomorphism on the fibers over U_V. Also, h is continuous because its components are continuous. It remains to verify that h^{-1} is continuous. Given $V' \in U_V$, there is a unique invertible map $L_{V'} : \mathbb{R}^n \to \mathbb{R}^n$

restricting to π_V on V' and the identity on $V^\perp = \ker \pi_V$. Regarding $L_{V'}$ as an $n \times n$ matrix, we assert that it varies continuously with respect to the subspace V'.

To demonstrate this, we construct a factorization $L_{V'} = AB^{-1}$ using a local frame. Let $\{u_1, \ldots, u_k\}$ be an orthonormal basis for V', and fix a constant basis $\{u_{k+1}, \ldots, u_n\}$ for the orthogonal complement $V^\perp$. We define B to be the change-of-basis matrix mapping the standard basis vector e_i to u_i for all $1 \leq i \leq n$. Similarly, we define A such that it sends e_i to $\pi_V(u_i)$ for $1 \leq i \leq k$, and acts as the identity on the complement by sending e_i to u_i for $i > k$. The matrices A and B are clearly continuous functions of the orthonormal frame $\{u_1, \ldots, u_k\}$, and consequently, the product AB^{-1} depends continuously on this frame. Crucially, the linear operator $L_{V'}$ is intrinsic to the subspace V' and is independent of the specific basis chosen. Since the topology on $G_k(\mathbb{R}^n)$ is the quotient topology induced by the map from orthonormal frames, it follows that the map $V' \mapsto L_{V'}$ is continuous.

We can now express the inverse homeomorphism explicitly as $h^{-1}(V', w) = (V', L_{V'}^{-1} w)$. Since matrix inversion is a continuous operation on the general linear group $GL_n(\mathbb{R})$, the dependence of $L_{V'}^{-1}$ on V' is continuous, and thus h^{-1} is continuous.

This completes the proof for finite n. For the infinite case $n = \infty$, we construct the trivializing neighborhood U_V as the union of the sets defined for each finite dimension. The local trivializations constructed above are compatible under inclusion and glue together to define a homeomorphism on the direct limit. The continuity of this map is guaranteed by the definition of the direct limit topology on $G_k(\mathbb{R}^\infty)$. $\square$

2.3 Isomorphisms of Vector Bundles

Having explored various examples of vector bundles, a natural question arises: How many distinct vector bundles exist over a given space? However, the term "distinct" currently lacks a rigorous definition; to answer this, we must clarify what it means for two bundles to be the same, which necessitates a notion of isomorphism. Before defining isomorphisms, we first establish the general concept of maps between vector bundles, vector bundle morphisms, which will be needed for stating later results more compactly.

Definition 2.9. Let $p : E \to B$ and $p' : E' \to B'$ be vector bundles. A vector bundle morphism is a pair of maps $f' : E \to E'$ and $f : B \to B'$ such that the diagram

$$\begin{tikzcd} E \arrow[r, "f'"] \arrow[d, "p"] & E' \arrow[d, "p'"] \\ B \arrow[r, "f"] & B' \end{tikzcd} \tag{2.1}$$

commutes and the restriction $f'|_{p^{-1}(b)} : p^{-1}(b) \to (p')^{-1}(f(b))$ is linear for each $b \in B$.

When the total and base spaces are clear from context, we will denote the morphism simply as the pair (f', f). Note that the commutativity condition $p' \circ f' = f \circ p$ is equivalent to requiring that $f'(p^{-1}(b)) \subseteq (p')^{-1}(f(b))$ for each $b \in B$. This ensures that the map f' takes the fiber over b to the fiber over $f(b)$.

It is worth noting that the map f is uniquely determined by f'. To see this, observe that since p is surjective, every $b \in B$ has a non-empty fiber $p^{-1}(b)$. For any $e \in p^{-1}(b)$, the commutativity condition implies $f(b) = p'(f'(e))$. This definition is well-defined because f' maps the entire fiber $p^{-1}(b)$ into the fiber over $f(b)$, meaning $p'(f'(e))$ yields the same value $f(b)$ regardless of the choice of $e \in p^{-1}(b)$.

Remark 2.10. We will be particularly interested in morphisms between bundles sharing the same base space B, in which case the map f is taken to be the identity and the morphism (f', id_B) is referred to as a B-morphism. In this case, the commutativity condition simplifies to $p' \circ f' = p$, and the commutative diagram becomes triangular:

$$\begin{tikzcd} E \arrow[rr, "f'"] \arrow[dr, "p"'] & & E' \arrow[dl, "p'"] \\ & B & \end{tikzcd}$$

We can now give the definition of isomorphic bundles over the same base space.

Definition 2.11. Let (f', id_B) be a B-morphism between $\xi = (E, p, B)$ and $\xi' = (E', p', B)$. We say that (f', id_B) is a vector bundle isomorphism or a B-isomorphism, if $f' : E \to E'$ is a homeomorphism and its restriction to each fiber $f'|_{p^{-1}(b)} : p^{-1}(b) \to (p')^{-1}(b)$ is a vector space isomorphism.

In particular, isomorphic vector bundles have the same dimension.

Remark 2.12. There is a more general notion of isomorphism between bundles over different base spaces, but we will not need it for our discussion.

One crucial reason for studying isomorphisms is that they allow us to organize the otherwise unmanageable collection of all vector bundles into distinct equivalence classes, as the next result shows.

Proposition 2.13. The relation of being isomorphic, denoted by $\cong$, is an equivalence relation on the collection of vector bundles over a fixed base space B.

Proof. Reflexivity and symmetry are simple consequences of the properties of identity and inverse maps. We check transitivity. Suppose $\xi \cong \xi'$ and $\xi' \cong \xi''$, with isomorphisms $f' : E \to E'$ and $g' : E' \to E''$. The composition $g' \circ f' : E \to E''$ is a homeomorphism since it is a composition of homeomorphisms. Furthermore, for any $b \in B$, the restriction $(g' \circ f')|_{p^{-1}(b)}$ is the composition of two linear isomorphisms, which is itself a linear isomorphism. Thus, $(g' \circ f', \mathrm{id}_B)$ is an isomorphism, so $\xi \cong \xi''$. $\square$

We will denote the set of isomorphism classes of k-dimensional vector bundles over a space B by $\mathrm{Vect}_k(B)$.

Let us make the definitions we have seen more concrete with a classic example.

Example 2.14. Consider the tangent bundle of the circle S^1, denoted by $p : TS^1 \to S^1$. We regard S^1 as the subset of unit complex numbers in $\mathbb{C} \cong \mathbb{R}^2$. For any point $x \in S^1$, the tangent space $T_x S^1$ consists of vectors orthogonal to the position vector x. We claim that TS^1 is isomorphic to the trivial bundle $\epsilon^1 = S^1 \times \mathbb{R}$. To see this, observe that for any $x \in S^1$, the vector ix corresponding to a rotation of x by $\pi/2$ is tangent to the circle at x and nonzero. We can therefore define a map $f' : S^1 \times \mathbb{R} \to TS^1$ by $f'(x,t) = (x, t \cdot ix)$.

This map is clearly continuous as it is defined by standard complex multiplication. Moreover, it is a B-morphism because it preserves the base point x, i.e., $p \circ f' = \mathrm{pr}_1$. Crucially, for each fixed x, the restriction to the fiber maps $t \mapsto t \cdot ix$ is a linear isomorphism from $\mathbb{R}$ to the one-dimensional subspace $T_x S^1$. Since f' is a continuous bijection that is a linear isomorphism on fibers, it is a vector bundle isomorphism (see Lemma 2.15). Thus, the tangent bundle of the circle is trivial, $TS^1 \cong S^1 \times \mathbb{R}$.

It turns out that there is a much simpler condition to check whether a B-morphism is an isomorphism, as the next lemma shows:

Lemma 2.15. A B-morphism (f', id_B) is an isomorphism if and only if its restriction to each fiber $f'|_{p^{-1}(b)} : p^{-1}(b) \to (p')^{-1}(b)$ is a linear isomorphism.

Proof. We first check that diagram (2.1) is satisfied. Let $e \in p^{-1}(b)$. Then,

$$(p' \circ f')(e) = b = (\mathrm{id}_B \circ p)(e),$$

where the first equality follows since we are given that f' takes the fiber $p^{-1}(b)$ to $(p')^{-1}(b)$.

Next, we show that f' is bijective. Suppose that $f'(e_1) = f'(e_2)$ for some $e_1, e_2 \in E$. Applying p' on the left, we get

$$p(e_1) = (p' \circ f')(e_1) = (p' \circ f')(e_2) = p(e_2).$$

Let $b = p(e_1) = p(e_2)$. Then e_1, e_2 lie in the same fiber $p^{-1}(b)$. Since f' restricted to this fiber is a linear isomorphism (and thus injective), it follows that $e_1 = e_2$. To show surjectivity, let $e' \in E'$. Let $b = p'(e')$. Since f' maps $p^{-1}(b)$ surjectively onto $(p')^{-1}(b)$, there exists $e \in p^{-1}(b)$ such that $f'(e) = e'$. Thus f' is bijective.

Finally, we show that $(f')^{-1}$ is continuous. Since continuity is a local property, it suffices to show $(f')^{-1}$ is continuous on a neighborhood of any point. Given $b_0 \in B$, choose an open neighborhood $U \subseteq B$ of b_0 over which both E and E' are trivial and let $h : p^{-1}(U) \to U \times \mathbb{R}^k$ and $h' : (p')^{-1}(U) \to U \times \mathbb{R}^k$ be the corresponding local trivializations. Consider the composite map $F = h' \circ f' \circ h^{-1}$ defined on $U \times \mathbb{R}^k$. Since f' is linear on fibers, F must have the form

$$F(b, v) = (b, g(b)v),$$

where $g(b)$ is an invertible $k \times k$ matrix for each $b \in U$. The map $b \mapsto g(b)$ from U to $GL_k(\mathbb{R})$ is continuous because f' (and thus F) is continuous. The inverse map F^{-1} is given by

$$F^{-1}(b, v) = (b, g(b)^{-1}v).$$

Since matrix inversion is a continuous operation on $GL_k(\mathbb{R})$, the map $b \mapsto (g(b))^{-1}$ is continuous, which implies F^{-1} is continuous. Since $(f')^{-1} = h^{-1} \circ F^{-1} \circ h'$ on $(p')^{-1}(U)$, it follows that $(f')^{-1}$ is continuous. $\square$

2.4 Induced Bundles

In this subsection, we introduce a tool of crucial importance that will be useful when we classify vector bundles. The way this tool arises comes from considering the following question:

> Given a vector bundle $\xi = (E, p, B)$ and a map $f : B' \to B$, is it possible to define a vector bundle over B' whose structure is dictated by the original bundle ξ and the map f?

The answer to this question is affirmative. Moreover, the resulting object is unique and known as the bundle induced by f or the pullback of ξ by f, denoted $f^*(\xi)$. This construction is the fundamental mechanism that links the geometry of ξ over B to a new bundle over B', preserving the fiber structure as dictated by the map f.

Let us now work out the details. Denote[4] the total space of the induced bundle by E' or $f^*(E)$, which is formally defined as the fiber product of the spaces B' and E over B:

$$E' = \{(b', e) \in B' \times E \mid f(b') = p(e)\}.$$

This ensures that the fiber over any point $b' \in B'$ is precisely the fiber of the original bundle E over the corresponding point $f(b') \in B$. Namely, the projection $p' : E' \to B'$ of the pullback bundle is given by

$$p'(b', e) = b'.$$

It can be easily checked that the induced bundle $f^*(\xi)$ satisfies the following commuting diagram:

$$\begin{tikzcd} E' \arrow[r, "\mathrm{pr}_2"] \arrow[d, "p'"'] & E \arrow[d, "p"] \\ B' \arrow[r, "f"'] & B \end{tikzcd} \tag{2.2}$$

[4]From this point forward, the reader is encouraged to maintain a separate log of the notation employed. Due to the density of the material, not every piece of shorthand is formally introduced, but rather is intended to be self-evident from the immediate context or prior usage. This reliance on context is a shared necessity of algebraic topology, where even the author may occasionally struggle to keep track of the complete notational history across every section.

In this diagram, pr_2 is the projection onto the second factor. Here is the main result of this subsection:

Proposition 2.16. Given a vector bundle $p : E \to B$ and a continuous map $f : B' \to B$, the induced bundle $p' : E' \to B'$ admits the structure of a vector bundle such that the map $\tilde{f} : E' \to E$ defined by $\tilde{f}(b', e) = e$ gives a morphism $(\tilde{f}, f)$ of vector bundles. Moreover, such a bundle, denoted by $f^*(\xi)$, is unique up to isomorphism.

Proof. Recall that E' is defined as the subset $E' = \{(b', e) \in B' \times E \mid f(b') = p(e)\} \subseteq B' \times E$, with $p' : E' \to B'$ being the projection onto the first factor.

First, note that the fiber over b' is given by $(p')^{-1}(b') = \{b'\} \times p^{-1}(f(b'))$. Since $p^{-1}(f(b'))$ is a vector space, this suggests we define the vector space operations on $(p')^{-1}(b')$ as follows:

$$(b', e_1) + (b', e_2) = (b', e_1 + e_2) \quad \text{and} \quad c \cdot (b', e) = (b', c \cdot e)$$

for $(b', e_1), (b', e_2) \in (p')^{-1}(b')$ and scalar c. One easily checks that these operations turn the fiber into a vector space isomorphic to $p^{-1}(f(b'))$.

It remains to show that E' is locally trivial. Let $b'_0 \in B'$ and let $b_0 = f(b'_0) \in B$. Since E is a vector bundle, there exists an open neighborhood $U \subseteq B$ of b_0 and a local trivialization $\phi : p^{-1}(U) \to U \times \mathbb{R}^k$.

Define $U' = f^{-1}(U)$, which is an open neighborhood of b'_0 in B'. We construct a local trivialization $\Phi' : (p')^{-1}(U') \to U' \times \mathbb{R}^k$ by setting:

$$\Phi'(b', e) = (b', \pi_2(\phi(e))),$$

where $\pi_2 : U \times \mathbb{R}^k \to \mathbb{R}^k$ is the projection onto the vector component. This map is continuous because f, ϕ, and the projection are continuous. Its inverse is given by:

$$(\Phi')^{-1}(b', v) = (b', \phi^{-1}(f(b'), v)),$$

which is also continuous. Furthermore, for a fixed b', the map $e \mapsto \pi_2(\phi(e))$ is a linear isomorphism by the properties of ϕ. Thus, Φ' is a valid local trivialization. Since B' is covered by such neighborhoods, E' is a vector bundle.

Finally, the map $\tilde{f} : E' \to E$ defined by $\tilde{f}(b', e) = e$ is clearly continuous (it is the projection from $B' \times E$) and linear on fibers, satisfying the definition of a morphism. Uniqueness up to isomorphism follows immediately from Lemma 2.15: if another bundle E'' satisfies these properties, the fiberwise isomorphisms glue together to form a global isomorphism. □

Theorem 2.17. Let $\xi_1 = (E_1, p_1, B_1)$ and $\xi_2 = (E_2, p_2, B_2)$ be two vector bundles. Given a map $f : B_2 \to B_1$, the bundles ξ_2 and $f^*(\xi_1)$ are isomorphic if and only if there exists a morphism $(F, f) : \xi_2 \to \xi_1$ such that F is a linear isomorphism on fibers of ξ_2.

Proof. Suppose $\xi_2 \cong f^*(\xi_1)$. Let $\Phi : E_2 \to f^*(E_1)$ be a bundle isomorphism. Recall that the pullback bundle comes with a natural morphism $(\tilde{f}, f) : f^*(\xi_1) \to \xi_1$ defined by $\tilde{f}(b_2, e_1) = e_1$. We define the map $F = \tilde{f} \circ \Phi : E_2 \to E_1$. Since both Φ and $\tilde{f}$ are linear isomorphisms on fibers (the latter by definition of the pullback vector space structure), their composition F is a linear isomorphism on fibers. Thus, (F, f) is the desired morphism.

Conversely, suppose there exists a morphism $(F, f) : \xi_2 \to \xi_1$ such that F restricts to a linear isomorphism on each fiber. We define a map $\Psi : E_2 \to f^*(E_1)$ by

$$\Psi(e) = (p_2(e), F(e)).$$

First, we verify that the image lies in $f^*(E_1)$. By definition, $f^*(E_1) = \{(b_2, e_1) \in B_2 \times E_1 \mid f(b_2) = p_1(e_1)\}$. For any $e \in E_2$, we have $p_1(F(e)) = f(p_2(e))$ because (F, f) is a bundle morphism. Thus, Ψ is well-defined.

The map Ψ is continuous because its components p_2 and F are continuous. Moreover, for any $b \in B_2$, Ψ maps the fiber $p_2^{-1}(b)$ to the fiber of $f^*(\xi_1)$ over b, which is $\{b\} \times p_1^{-1}(f(b))$. The map on this fiber is given by $v \mapsto (b, F(v))$. Since F is a linear isomorphism on fibers, Ψ is a linear isomorphism on fibers.

Therefore, Ψ is a morphism that is an isomorphism on fibers. By Lemma 2.15, Ψ is a vector bundle isomorphism, so $\xi_2 \cong f^*(\xi_1)$. □

2.5 Classification of Vector Bundles

With the tools we have developed so far, we can now state the main result of this project. Recall from homotopy theory that $[X, Y]$ denotes the set of homotopy classes of maps from X to Y. The culmination of our work is the following classification theorem:

Theorem 2.18. Let B be a paracompact space. The function

$$\Theta : [B, G_k(\mathbb{R}^\infty)] \to \mathrm{Vect}_k(B)$$

given by $\Theta([f]) = [f^*(\gamma_k)]$ is a bijection.

Proof. The map Θ is well-defined by Theorem 2.24, surjective by Corollary 2.28, and injective by Theorem 2.34. $\square$

This bijection allows us to translate the geometric problem of classifying vector bundles into the homotopy-theoretic problem of classifying maps:

Corollary 2.19. Let B be a paracompact space. Two vector bundles over B are isomorphic if and only if their classifying maps are homotopic.

Proof. Let ξ and ξ' be vector bundles classified by maps $f_1, f_2 : B \to G_k(\mathbb{R}^\infty)$ respectively.

If $\xi \cong \xi'$, then $[\xi] = [\xi']$ in $\mathrm{Vect}_k(B)$. Since Θ is a bijection (specifically, injective), we have:

$$\Theta([f_1]) = \Theta([f_2]) \implies [f_1] = [f_2].$$

Thus, $f_1 \simeq f_2$.

Conversely, if $f_1 \simeq f_2$, then $[f_1] = [f_2]$. Since Θ is a well-defined function,

$$\Theta([f_1]) = \Theta([f_2]) \implies [f_1^*(\gamma_k)] = [f_2^*(\gamma_k)].$$

Therefore, the bundles are isomorphic, $\xi \cong \xi'$. $\square$

The remainder of this report is dedicated to establishing the fundamental results listed in the proof of Theorem 2.18.

2.6 Homotopy Invariance

We now begin discussing some technical results involving the homotopy properties of pullback bundles introduced in the previous subsection. As the results are technical, they require a technical assumption about the base space, namely that it be paracompact (for a treatment of paracompact spaces, see [Lee11]). Nevertheless, the results we establish are quite general (for example, manifolds and CW complexes are paracompact, see [Hat01] and [Lee11]) and are the first step in classifying vector bundles. We start with some lemmata:

Lemma 2.20. Let $p : E \to B \times [a, b]$ be a vector bundle such that its restrictions over $B \times [a, c]$ and $B \times [c, b]$ are both trivial for some $c \in (a, b)$. Then $p : E \to B \times [a, b]$ is trivial.

Proof. Let E_1 and E_2 denote the restrictions of E to $B \times [a, c]$ and $B \times [c, b]$ respectively. By hypothesis, there exist trivializations:

$$\phi_1 : E_1 \to (B \times [a, c]) \times \mathbb{R}^k \quad \text{and} \quad \phi_2 : E_2 \to (B \times [c, b]) \times \mathbb{R}^k.$$

We restrict our attention to the intersection $B \times \{c\}$. For any $b \in B$ and $v \in \mathbb{R}^k$, the composition $\phi_2 \circ \phi_1^{-1}$ restricts to an isomorphism of the fiber over (b, c). Thus, we can write:

$$\phi_2(\phi_1^{-1}(b, c, v)) = (b, c, g(b)v),$$

where $g : B \to GL_k(\mathbb{R})$ is a continuous map describing the transition between the two trivializations at the interface $t = c$.

To glue these into a global trivialization, we need to modify ϕ_2 so that it agrees with ϕ_1 at c. Define a new map $\tilde{\phi}_2 : E_2 \to (B \times [c, b]) \times \mathbb{R}^k$ by

$$\tilde{\phi}_2(e) = (b, t, g(b)^{-1}v'), \quad \text{where } \phi_2(e) = (b, t, v').$$

Note that this uses the projection $B \times [c, b] \to B$ to apply $g(b)^{-1}$ at each $t \in [c, b]$. Since g is continuous and takes values in invertible matrices, $\tilde{\phi}_2$ is a valid vector bundle isomorphism (a composition of the isomorphism ϕ_2 and a linear isomorphism).

Now we check the boundary condition at $t = c$. For $e \in p^{-1}(b, c)$, let $\phi_1(e) = (b, c, v)$. Then by our definition of g, we have $\phi_2(e) = (b, c, g(b)v)$. Applying our modified map, we have

$$\tilde{\phi}_2(e) = (b, c, g(b)^{-1}(g(b)v)) = (b, c, v) = \phi_1(e).$$

Since ϕ_1 and $\tilde{\phi}_2$ agree on $E_1 \cap E_2$, we get a global map $\Phi : E \to B \times [a, b] \times \mathbb{R}^k$. As Φ is a linear isomorphism on each fiber and a homeomorphism on the total space, E is trivial. $\square$

Lemma 2.21. Let $\xi = (E, p, B \times I)$ be a vector bundle, where I is the closed unit interval. Then, there exists an open cover $\{U_\alpha\}_{\alpha \in A}$ of B, where A is some index set, such that $\xi|(U_\alpha \times I)$ is trivial.

Proof. By the local triviality of ξ, for each $t \in I$, there exist open neighborhoods U_t of b in B and V_t of t in I such that the restricted bundle $\xi|_{U_t \times V_t}$ is trivial.

The collection $\{V_t\}_{t \in I}$ is an open cover of I. By the compactness of I, this cover reduces to a finite subcover, say $\{V_{t_j}\}_{j=1}^k$. For each $j \in \{1, \dots, k\}$, the bundle $\xi|_{U_{t_j} \times V_{t_j}}$ is trivial.

By the Lebesgue Number Lemma (e.g., [Lee11]), there exists a partition $0 = t_0 < t_1 < \cdots < t_K = 1$, such that each subinterval $I_i = [t_{i-1}, t_i]$ is entirely contained in one of the open sets V_{t_j} from the finite subcover.

Let $U_i \subseteq B$ denote the open set U_{t_j} corresponding to the covering set V_{t_j} containing I_i. Since $I_i \subseteq V_{t_j}$, it follows that $U_i \times I_i \subseteq U_{t_j} \times V_{t_j}$, and thus $\xi|_{U_i \times I_i}$ is trivial for all $i \in \{1, \dots, K\}$.

Now, let $U = \bigcap_{i=1}^K U_i$. Since U is a finite intersection of open sets containing b, it is an open neighborhood of b. Since $U \subseteq U_i$ for all i, the restricted bundle $\xi|_{U \times I_i}$ is trivial. The base space $U \times I$ is the union of the closed sets $U \times I_i$. By applying Lemma 2.20 $K - 1$ times to join the trivial bundles $\xi|_{U \times I_i}$ and $\xi|_{U \times I_{i+1}}$ over the common intersection $U \times \{t_i\}$, we conclude that $\xi|_{U \times I}$ is trivial.

Since $b \in B$ was arbitrary, the collection of all such open sets U forms the desired open cover $\{U_\alpha\}_{\alpha \in A}$ of B. □

The following theorem will lead us to the first important result for classifying bundles:

Theorem 2.22. For a paracompact space B, let $r : B \times I \to B \times I$ be given by $r(b, t) = (b, 1)$, and let $p : E \to B \times I$ be a vector bundle. Then there exists a map $u : E \to E$ such that the diagram

$$\begin{tikzcd} E \arrow[r, "u"] \arrow[d, "p"] & E \arrow[d, "p"] \\ B \times I \arrow[r, "r"] & B \times I \end{tikzcd} \tag{2.3}$$

commutes and u takes $p^{-1}(b, t)$ onto $p^{-1}(r(b, t))$ by a linear isomorphism.

Proof. By the paracompactness of B and Lemma 2.21, we have a locally finite open covering $\{U_\alpha\}_{\alpha\in A}$ such that $\xi|(U_\alpha \times I)$ is trivial, since the restriction of a trivial bundle is again trivial. Let $\{\psi_\alpha\}_{\alpha\in A}$ be a partition of unity subordinate to $\{U_\alpha\}_{\alpha\in A}$, and let $h_\alpha : U_\alpha \times I \times \mathbb{R}^k \to p^{-1}(U_\alpha \times I)$ be a local trivialization. Define $u_\alpha : E \to E$ and $r_\alpha : B \times I \to B \times I$ by

$$r_\alpha(b,t) = (b, \max(\psi_\alpha(b), t))$$

$$u_\alpha(e) = \begin{cases} e, & \text{if } e \notin p^{-1}(U_\alpha \times I) \\ h_\alpha(b, \max(\psi_\alpha(b), t), x), & \text{if } h_\alpha(b,t,x) = e \in p^{-1}(U_\alpha \times I). \end{cases}$$

Let us check $r_\alpha \circ p = p \circ u_\alpha$. For $e \notin p^{-1}(U_\alpha \times I)$, let $p(e) = (b,t)$. Then, $b \notin U_\alpha$ as $p(e) \notin U_\alpha \times I$. Since $\operatorname{supp}(\psi_\alpha) \subseteq U_\alpha$, we have $\psi_\alpha(b) = 0$ and thus

$$(r_\alpha \circ p)(e) = r_\alpha(b,t) = (b, \max(\psi_\alpha(b), t)) = (b,t) = p(e).$$

For $h_\alpha(b,t,x) = e \in p^{-1}(U_\alpha \times I)$, we have

$$p(u_\alpha(e)) = p(h_\alpha(b, \max(\psi_\alpha(b), t), x)) = \mathrm{pr}_1(b, \max(\psi_\alpha(b), t), x) = (b, \max(\psi_\alpha(b), t)),$$

where we used the definition of the trivialization to write $p \circ h_\alpha$ as the projection to the base. On the other hand, we have

$$r_\alpha(p(e)) = r_\alpha(p(h_\alpha(b,t,x))) = r_\alpha(\mathrm{pr}_1(b,t,x)) = r_\alpha(b,t) = (b, \max(\psi_\alpha(b), t)).$$

Thus, in both cases, we have $r_\alpha \circ p = p \circ u_\alpha$.

We now construct the global map u by composing the maps u_α. Since the cover $\{U_\alpha\}$ is locally finite, we can impose a well-ordering on the index set A. For any point $b \in B$, there exists an open neighborhood $W_b \subseteq B$ that intersects only finitely many elements of the cover, say $U_{\alpha_1}, \ldots, U_{\alpha_k}$ with indices $\alpha_1 < \alpha_2 < \cdots < \alpha_k$. Consequently, for any $e \in p^{-1}(W_b \times I)$, all but finitely many of the maps u_α act as the identity. This allows us to meaningfully define the global maps r and u as the ordered infinite compositions:

$$r = \bigcirc_{\alpha\in A} r_\alpha \quad \text{and} \quad u = \bigcirc_{\alpha\in A} u_\alpha.$$

Restricted to the neighborhood $W_b \times I$, these collapse to the finite compositions $r = r_{\alpha_k} \circ \cdots \circ r_{\alpha_1}$ and $u = u_{\alpha_k} \circ \cdots \circ u_{\alpha_1}$. Since finite compositions of continuous maps are continuous, r and u are well-defined and continuous globally.

It remains to show that u is an isomorphism on each fiber. Since u is a composition of maps u_α, and each u_α is a linear isomorphism on fibers (being a change of coordinates via local trivializations), the composition is necessarily a linear isomorphism on each fiber. □

Here is the most important corollary of the theorem we just proved:

Corollary 2.23. For a vector bundle $p : E \to B \times I$ with B paracompact, the restrictions over $B \times \{a\}$ and $B \times \{b\}$ are isomorphic for $a, b \in I$ with $a < b$.

Proof. Let $r : B \times I \to B \times I$ be the retraction onto the level $t = b$, defined by $r(x, t) = (x, b)$. By Theorem 2.22, there exists a vector bundle map (u, r), where $u : E \to E$ is a linear isomorphism on each fiber, such that the following diagram commutes:

$$\begin{tikzcd}
E \arrow[r, "u"] \arrow[d, "p"] & E \arrow[d, "p"] \\
B \times I \arrow[r, "r"] & B \times I
\end{tikzcd}$$

We restrict the map u to the total space of the bundle over the slice $t = a$, denoted by $E_a = p^{-1}(B \times \{a\})$. Let $u_a = u|_{E_a}$. For any vector e in the fiber over (x, a), the commutativity condition $p \circ u = r \circ p$ implies that $p(u(e)) = r(x, a) = (x, b)$. Consequently, u_a maps the fiber over (x, a) into the fiber over (x, b), effectively defining a map between total spaces $u_a : E_a \to E_b$ covering the natural identification $B \times \{a\} \to B \times \{b\}$.

Crucially, the global map u is constructed to be a linear isomorphism on every fiber of E. Therefore, its restriction to any specific fiber $p^{-1}(x, a)$ is necessarily a linear isomorphism onto $p^{-1}(x, b)$. Since u_a is a continuous map covering restricting to a linear isomorphism on fibers, it gives a vector bundle isomorphism. Thus, $E|_{B \times \{a\}} \cong E|_{B \times \{b\}}$. □

Here is the first main result:

Theorem 2.24. Let $f_0, f_1 : B \to B'$ be homotopic maps with B paracompact, and let $\xi = (E, p, B')$ be a vector bundle. Then the induced bundles $f_0^*(\xi)$ and $f_1^*(\xi)$ are isomorphic.

Proof. Let $H : B \times I \to B'$ be a homotopy from f_0 to f_1, so that $H(b, 0) = f_0(b)$ and $H(b, 1) = f_1(b)$. Consider the induced bundle over the product space, $H^*(\xi)$, which is a vector bundle over $B \times I$.

By Corollary 2.23, the restrictions of this bundle to the slices $B \times \{0\}$ and $B \times \{1\}$ are isomorphic:

$$H^*(\xi)|_{B\times\{0\}} \cong H^*(\xi)|_{B\times\{1\}}.$$

It suffices to show that restriction to the slice $t = 0$ recovers the pullback by f_0. The total space of $H^*(\xi)|_{B\times\{0\}}$ is defined as:

$$E|_{B\times\{0\}} = \{((b, 0), e) \in (B \times \{0\}) \times E \mid H(b, 0) = p(e)\}.$$

Since $H(b, 0) = f_0(b)$, the condition becomes $f_0(b) = p(e)$. There is a map $\phi : f_0^*(\xi) \to H^*(\xi)|_{B\times\{0\}}$ given by $\phi(b, e) = ((b, 0), e)$. This map is clearly a homeomorphism on total spaces and acts as the identity on the vector component e, preserving the linear structure of the fibers. Thus, ϕ is an isomorphism of vector bundles.

Applying the same logic to $t = 1$, we obtain the chain of isomorphisms:

$$f_0^*(\xi) \cong H^*(\xi)|_{B\times\{0\}} \cong H^*(\xi)|_{B\times\{1\}} \cong f_1^*(\xi).$$

This completes the proof. □

2.7 Gauss Maps

Having established that homotopic maps induce isomorphic vector bundles, we now turn to the surjectivity of Θ. Our goal is to prove that every rank-k vector bundle ξ is isomorphic to the pullback of the universal bundle γ_k via a map $f : B \to G_k(\mathbb{R}^\infty)$. Let us start with a definition:

Definition 2.25. Let $\xi = (E, p, B)$ be a k-dimensional vector bundle. A map $g : E \to \mathbb{R}^m$ is called a Gauss map of ξ in $\mathbb{R}^m$ ($k \leq m \leq \infty$) if its restriction to each fiber of ξ is a linear injection.

More explicitly, g is a Gauss map if it maps each fiber of ξ to a k-dimensional subspace of $\mathbb{R}^m$.

Example 2.26. Recall from the definition of the canonical bundle that $E_k(\mathbb{R}^n) = \{(V,x) \in G_k(\mathbb{R}^n) \times \mathbb{R}^n \mid x \in V\}$. Then the projection $q : E_k(\mathbb{R}^n) \to \mathbb{R}^n$ given by $q(V,x) = x$ is easily seen to be a Gauss map by the following simple calculation:

$$\begin{aligned} p^{-1}(V) &= \{(V',x) \in G_k(\mathbb{R}^n) \times \mathbb{R}^n \mid x \in V' \text{ and } p(V',x) = V' = V\} \\ &= \{(V',x) \in G_k(\mathbb{R}^n) \times \mathbb{R}^n \mid x \in V\} = \{V\} \times V \cong V. \end{aligned}$$

Restricted to this fiber, $q(V,x) = x$ acts as the identity on V (viewed as a subspace of $\mathbb{R}^n$), which is clearly a linear injection.

As in the previous sections, we now prove some technical results. The first of these results shows that the existence of the Gauss map q of Example 2.26 provides a link between vector bundle morphisms and Gauss maps. In the following, we have $k \leq n \leq \infty$.

Proposition 2.27. Let (F,f) be a vector bundle morphism between a rank-k bundle $\xi^k = (E,p,B)$ and γ_k^n such that F is an isomorphism on fibers of ξ^k. Then the composition $q \circ F : E \to \mathbb{R}^n$ is a Gauss map. Conversely, given a Gauss map $g : E \to \mathbb{R}^n$, there exists a vector bundle morphism (F,f) such that $q \circ F = g$.

Proof. Let $e_1, e_2 \in p^{-1}(b)$. Then

$$q(F(e_1+e_2)) = q(F(e_1)+F(e_2)) = q(F(e_1)) + q(F(e_2))$$

as $F(e_1), F(e_2) \in (p')^{-1}(f(b))$ and q restricts to a linear map on $(p')^{-1}(f(b))$. Similarly, $q(F(ce)) = q(cF(e)) = cq(F(e))$, showing linearity on fibers. To check injectivity, let $q(F(e_1)) = q(F(e_2))$ for some $e_1, e_2 \in p^{-1}(b)$. Since q is injective on the fibers of the canonical bundle (it is just the inclusion $V \hookrightarrow \mathbb{R}^n$), we have $F(e_1) = F(e_2)$. Since F is also injective on fibers by hypothesis, this implies $e_1 = e_2$.

Conversely, define $f : B \to G_k(\mathbb{R}^n)$ by sending b to the subspace of $\mathbb{R}^n$ that is the image of the fiber $p^{-1}(b)$ under g, i.e., $f(b) = g(p^{-1}(b))$. Then define $F : E \to E_k(\mathbb{R}^n)$ by $F(e) = (f(p(e)), g(e))$. Since $g(e) \in f(p(e))$ by construction, this map lands in the total space $E_k(\mathbb{R}^n)$.

The condition $p' \circ F = f \circ p$ holds immediately:

$$p'(F(e)) = p'(f(p(e)), g(e)) = f(p(e)).$$

To check the continuity of f, consider a local trivialization $h : U \times \mathbb{R}^k \to p^{-1}(U)$ of the bundle ξ. The composition $g \circ h : U \times \mathbb{R}^k \to \mathbb{R}^n$ is continuous. For a fixed $b \in U$, the map $v \mapsto (g \circ h)(b, v)$ is a linear injection from $\mathbb{R}^k$ to $\mathbb{R}^n$. We can represent this family of linear maps by a continuous function $M : U \to \mathbb{R}^{n \times k}$ (specifically into the space of rank-k matrices), such that $(g \circ h)(b, v) = M(b)v$. The subspace $f(b)$ is precisely the column space of the matrix $M(b)$. Since the map sending a full-rank matrix to its column space (a map from the Stiefel manifold to the Grassmannian) is continuous (it is the quotient map defining the Grassmannian topology), the composition $b \mapsto M(b) \mapsto \mathrm{Col}(M(b)) = f(b)$ is continuous. Consequently, F is continuous as both its components are continuous.

Next, we show that F is a linear isomorphism on each fiber. For $e_1, e_2 \in p^{-1}(b)$, we have

$$\begin{aligned} F(e_1 + e_2) &= (f(b), g(e_1 + e_2)) \\ &= (f(b), g(e_1) + g(e_2)) \\ &= (f(b), g(e_1)) + (f(b), g(e_2)) \\ &= F(e_1) + F(e_2), \end{aligned}$$

where the addition in the third line takes place in the fiber of γ_k^n over $f(b)$ (which is $\{f(b)\} \times f(b)$). Similarly, $F(ce) = cF(e)$. To show injectivity, suppose $F(e_1) = F(e_2)$. Then $(f(b), g(e_1)) = (f(b), g(e_2))$, which implies $g(e_1) = g(e_2)$. Since g is a Gauss map (linear injection on fibers), $e_1 = e_2$.

Surjectivity follows from dimension counting: F maps the fiber $p^{-1}(b)$ injectively into the fiber $(p')^{-1}(f(b))$. Both fibers are k-dimensional vector spaces, so an injective linear map must be surjective. Thus, F is a fiberwise isomorphism.

Finally, we verify that $q \circ F = g$. For each $e \in E$, one has $q(F(e)) = q(f(p(e)), g(e)) = g(e)$. □

Admittedly, the proof of this proposition is a bit meticulous, but it leads to a nice corollary:

Corollary 2.28. A vector bundle $p : E \to B$ has a Gauss map $g : E \to \mathbb{R}^n$ if and only if it is isomorphic to $f^*(\gamma_k^n)$ for some map $f : B \to G_k(\mathbb{R}^n)$.

Proof. For the forward direction, first apply Proposition 2.27, then Theorem 2.17.

For the reverse direction, first apply Theorem 2.17, then Proposition 2.27. □

We need to establish one final result before we can show the surjectivity of the classifying map:

Proposition 2.29. Let $\xi = (E, p, B)$ be a vector bundle with B paracompact such that $\xi|_{U_\alpha}$ is trivial for an open cover $\{U_\alpha\}_{\alpha \in A}$ of B. Then there exists a countable open cover $\{W_m\}_{m \in \mathbb{N}}$ of B such that $\xi|_{W_m}$ is trivial.

Proof. Since B is paracompact, let $\{\psi_\alpha\}_{\alpha \in A}$ be a partition of unity subordinate to $\{U_\alpha\}_{\alpha \in A}$. For $b \in B$, let $S(b) := \{\alpha \in A \mid \psi_\alpha(b) > 0\}$. Note that this is a finite set as the partition of unity is locally finite. For each finite subset $T \subseteq A$, define the set:

$$W(T) := \{b \in B \mid \psi_\alpha(b) > \psi_\beta(b) \text{ for all } \alpha \in T \text{ and } \beta \notin T\}.$$

We first show that $W(T)$ is an open subset of B. Define $f(b) := \max_{\beta \notin T} \psi_\beta(b)$. Since $S(b)$ is finite, for any point b there is a neighborhood where only finitely many ψ_β are nonzero. Thus, locally, f is the maximum of a finite number of continuous functions. Since the maximum of a finite collection of continuous functions is continuous (see e.g., [Mun00]), f is continuous on B. Observe that $b \in W(T)$ if and only if $\psi_\alpha(b) > f(b)$ for all $\alpha \in T$. Thus,

$$W(T) = \bigcap_{\alpha \in T} \{b \in B \mid \psi_\alpha(b) - f(b) > 0\}.$$

Since each set in the intersection is the preimage of $(0, \infty)$ under a continuous function, $W(T)$ is a finite intersection of open sets, hence open.

Next, we claim that if T and T' are distinct subsets with the same cardinality $|T| = |T'|$, then $W(T) \cap W(T') = \emptyset$. Since $T \neq T'$ and they have the same size, neither is contained in the other. Thus, there exists $\alpha \in T \setminus T'$ and $\beta \in T' \setminus T$. If $b \in W(T) \cap W(T')$, the definition of $W(T)$ implies $\psi_\alpha(b) > \psi_\beta(b)$ (since $\alpha \in T, \beta \notin T$), while the definition of $W(T')$ implies $\psi_\beta(b) > \psi_\alpha(b)$ (since $\beta \in T', \alpha \notin T'$). This is a contradiction, so the intersection is empty.

We can now construct the countable cover. For each integer $m \geq 1$, define

$$W_m = \bigcup_{|T|=m} W(T).$$

The collection $\{W_m\}_{m=1}^{\infty}$ covers B because for any b, if we take $T = S(b)$ and $m = |S(b)|$, then $b \in W(T) \subseteq W_m$. Since W_m is a disjoint union of open sets $W(T)$, we can define a trivialization on W_m by defining it separately on each component. For any T, if $b \in W(T)$, then for any $\alpha \in T$, we have $\psi_\alpha(b) > 0$, which implies $W(T) \subseteq \operatorname{supp}(\psi_\alpha) \subseteq U_\alpha$. Since $\xi|_{U_\alpha}$ is trivial, its restriction to $W(T)$ is trivial. Let $\phi_T : p^{-1}(W(T)) \to W(T) \times \mathbb{R}^k$ be such a trivialization. We define $\Phi_m : p^{-1}(W_m) \to W_m \times \mathbb{R}^k$ by setting $\Phi_m|_{p^{-1}(W(T))} = \phi_T$. This is well-defined and continuous due to the disjointness of the sets $W(T)$ for a fixed m. Thus, $\xi|_{W_m}$ is trivial for each m. $\square$

Remark 2.30. Observe that if each $b \in B$ is contained in at most n sets U_α, we can find a finite open cover $\{W_i\}_{i=1}^{n}$ of B such that $\xi|_{W_i}$ is trivial, since $S(b)$ can have cardinality at most n, implying W_m is empty for $m > n$.

We can now prove the main result about Gauss maps:

Theorem 2.31. Every vector bundle $p : E \to B$ over a paracompact space B has a Gauss map $g : E \to \mathbb{R}^\infty$.

Proof. By local triviality, we have an open cover $\{U_\alpha\}$ such that $\xi|_{U_\alpha}$ is trivial. Using Proposition 2.29, we refine this to a countable open cover $\{W_i\}$ such that $\xi|_{W_i}$ is trivial. Let $h_i : p^{-1}(W_i) \to W_i \times \mathbb{R}^k$ be a local trivialization and let $\{\psi_i\}$ be a partition of unity subordinate to $\{W_i\}$. Define $g : E \to \bigoplus_{i=1}^{\infty} \mathbb{R}^k \cong \mathbb{R}^\infty$ as $g(e) = (g_1(e), g_2(e), \dots)$, where

$$g_i(e) = \begin{cases} \psi_i(p(e)) \cdot (\mathrm{pr}_2 \circ h_i)(e), & \text{if } p(e) \in W_i \\ 0, & \text{otherwise.} \end{cases}$$

Each map g_i is continuous because the support of ψ_i is contained in W_i, ensuring the transition to zero outside W_i is smooth. Since the partition of unity is locally finite, every point $b \in B$ has a neighborhood intersecting only finitely many W_i. Consequently, the sum defining g is locally finite, implying that g is well-defined and continuous.

We now verify that g is a linear injection on each fiber. For a fixed $b \in B$, the restriction $g|_{p^{-1}(b)}$ is a linear map because each component g_i is simply a linear coordinate map scaled by the scalar $\psi_i(b)$.

To check injectivity, suppose $g(e) = 0$ for some vector $e \in p^{-1}(b)$. This implies that

$g_i(e) = 0$ for all i. By definition, this means:

$$\psi_i(b) \cdot (\mathrm{pr}_2 \circ h_i)(e) = 0 \quad \text{for all } i.$$

Since $\{\psi_i\}$ is a partition of unity, $\sum_i \psi_i(b) = 1$, so there must exist at least one index j such that $\psi_j(b) > 0$. For this index j, the equation above forces $(\mathrm{pr}_2 \circ h_j)(e) = 0$. Since h_j is a linear isomorphism on fibers, its projection onto the vector component is injective, which implies $e = 0$. Thus, g is injective on each fiber. □

We can finally show that the classifying map is surjective:

Corollary 2.32. Every vector bundle $p : E \to B$ over a paracompact space B is isomorphic to $f^*(\gamma_k)$ for some map $f : B \to G_k(\mathbb{R}^\infty)$.

Proof. We have a Gauss map $g : E \to \mathbb{R}^\infty$ by Theorem 2.31. Then Corollary 2.28 implies that ξ is isomorphic to $f^*(\gamma_k)$ for some map $f : B \to G_k(\mathbb{R}^\infty)$. □

2.8 Homotopy Properties of Gauss Maps

The final step in the classification is to show that Θ is injective. To do this, we need to show that if two maps f_1, f_2 induce isomorphic bundles, they are homotopic. The strategy involves embedding $\mathbb{R}^n$ into a larger space $\mathbb{R}^{2n}$ to separate the images of Gauss maps, allowing us to homotope between them without intersection.

Let $\mathbb{R}^{\text{even}}$ denote the subspace of vectors $x \in \mathbb{R}^\infty$ such that $x_{2i+1} = 0$ for $i \geq 0$ (i.e., odd-indexed coordinates are zero). Similarly, let $\mathbb{R}^{\text{odd}}$ be the subspace where $x_{2i} = 0$ for $i \geq 0$. Note that $\mathbb{R}^\infty = \mathbb{R}^{\text{even}} \oplus \mathbb{R}^{\text{odd}}$.

We define two homotopies of linear maps $g^{\text{even}} : \mathbb{R}^n \times I \to \mathbb{R}^{2n}$ and $g^{\text{odd}} : \mathbb{R}^n \times I \to \mathbb{R}^{2n}$ by the following formulas:

$$\begin{aligned} g_t^{\text{even}}(x_0, x_1, \dots) &= (1-t)(x_0, x_1, \dots) + t(x_0, 0, x_1, 0, \dots) \\ g_t^{\text{odd}}(x_0, x_1, \dots) &= (1-t)(x_0, x_1, \dots) + t(0, x_0, 0, x_1, \dots) \end{aligned}$$

where we identify $\mathbb{R}^n$ with the first n coordinates of $\mathbb{R}^{2n}$. The properties of these homotopies are summarized below.

Proposition 2.33. For $1 \leq n \leq \infty$, the maps defined above satisfy the following properties:

1. The maps g_0^{even} and g_0^{odd} each equal the standard inclusion $\iota : \mathbb{R}^n \to \mathbb{R}^{2n}$.

2. For $t = 1$, $g_1^{\text{even}}(\mathbb{R}^n) = \mathbb{R}^{2n} \cap \mathbb{R}^{\text{even}}$ and $g_1^{\text{odd}}(\mathbb{R}^n) = \mathbb{R}^{2n} \cap \mathbb{R}^{\text{odd}}$.

3. There exist bundle morphisms $(\tilde{f}^{\text{even}}, f^{\text{even}}) : \gamma_k^n \to \gamma_k^{2n}$ and $(\tilde{f}^{\text{odd}}, f^{\text{odd}}) : \gamma_k^n \to \gamma_k^{2n}$ such that
$$q_{2n} \circ \tilde{f}^{\text{even}} = g_1^{\text{even}} \circ q_n \quad \text{and} \quad q_{2n} \circ \tilde{f}^{\text{odd}} = g_1^{\text{odd}} \circ q_n.$$

4. The induced maps on the Grassmannian are homotopic to the inclusion: $f^{\text{even}} \simeq \iota$ and $f^{\text{odd}} \simeq \iota$ as maps $G_k(\mathbb{R}^n) \to G_k(\mathbb{R}^{2n})$.

Proof. The first two properties follow immediately from the formulas for g_t^{even} and g_t^{odd}. To establish the third property, we observe that the map g_1^{even} is a linear injection; therefore, its composition with the projection q_n constitutes a Gauss map into $\mathbb{R}^{2n}$. By applying Proposition 2.27, this Gauss map induces the required bundle morphism $(\tilde{f}^{\text{even}}, f^{\text{even}})$, and an identical argument applies to the odd case. Finally, the fourth property holds because the maps g_t^{even} and g_t^{odd} are continuous families of linear injections. These families naturally define homotopies of the induced maps on the Grassmannian, connecting f^{even} (and f^{odd}) to the map induced by g_0, which is simply the standard inclusion ι. $\square$

The following theorem establishes the injectivity of the classifying map ($k \leq n \leq \infty$ in the following).

Theorem 2.34. Let $f_1, f_2 : B \to G_k(\mathbb{R}^n)$ be two maps such that the induced bundles $f_1^*(\gamma_k^n)$ and $f_2^*(\gamma_k^n)$ are isomorphic. Let $\iota : G_k(\mathbb{R}^n) \to G_k(\mathbb{R}^{2n})$ be the natural inclusion. Then $\iota \circ f_1 \simeq \iota \circ f_2$.

Proof. Let $\xi = f_1^*(\gamma_k^n)$. By hypothesis, $\xi \cong f_2^*(\gamma_k^n)$, so we can view f_1 and f_2 as classifying maps for the same bundle ξ. By Corollary 2.28, these correspond to Gauss maps $g_1, g_2 : E \to \mathbb{R}^n$. We effectively need to show that the Gauss maps $\iota \circ g_1$ and $\iota \circ g_2$ into $\mathbb{R}^{2n}$ are homotopic through linear injections.

We lift these to $\mathbb{R}^{2n}$ and apply the homotopies from Proposition 2.33. By property (iv) of that proposition, $\iota \circ f_1$ is homotopic to $f^{\text{odd}} \circ f_1$. This latter map corresponds to the Gauss map $g_1^{\text{odd}} \circ g_1$, whose image lies entirely in the subspace $\mathbb{R}^{\text{odd}}$. Similarly, $\iota \circ f_2$

is homotopic to $f^{\text{even}} \circ f_2$, which corresponds to the Gauss map $g_1^{\text{even}} \circ g_2$, whose image lies entirely in $\mathbb{R}^{\text{even}}$.

We can now construct a linear homotopy between these two resulting maps. Define $H_t(e) = (1-t)(g_1^{\text{odd}} \circ g_1)(e) + t(g_1^{\text{even}} \circ g_2)(e)$. For any non-zero vector e, the vectors $(g_1^{\text{odd}} \circ g_1)(e)$ and $(g_1^{\text{even}} \circ g_2)(e)$ are non-zero vectors lying in the subspaces $\mathbb{R}^{\text{odd}}$ and $\mathbb{R}^{\text{even}}$ respectively. Since $\mathbb{R}^{\text{odd}} \cap \mathbb{R}^{\text{even}} = \{0\}$, these vectors are linearly independent, ensuring that $H_t(e)$ is never zero for any $t \in [0,1]$. Thus, H_t is a family of injective linear maps, which induces a homotopy between $f^{\text{odd}} \circ f_1$ and $f^{\text{even}} \circ f_2$. Combining these relations establishes that $\iota \circ f_1 \simeq \iota \circ f_2$. □

3 Conclusion

In this project, we have developed the classification theory of vector bundles, establishing the bridge between geometric bundles and homotopy theory. By constructing the infinite Grassmannian $G_k(\mathbb{R}^\infty)$ and proving the existence of Gauss maps, we have demonstrated the main classification theorem: for a paracompact base space B, the set of isomorphism classes of k-dimeansional vector bundles is in bijective correspondence with the set of homotopy classes of maps $[B, G_k(\mathbb{R}^\infty)]$.

This result is the fundamental theorem that underpins the modern topological study of vector bundles. It reduces the local geometric problem of patching trivializations to the global topological problem of understanding the homotopy type of the Grassmannian.

The significance of this classification lies in its power to define invariants. Because every vector bundle is a pullback of the universal bundle, any cohomology class of the infinite Grassmannian pulls back to define a characteristic class for vector bundles over B. While the calculation of specific classes, such as Stiefel-Whitney or Chern classes, depends on the cohomology ring of the Grassmannian, their existence, uniqueness, and well-definedness are guaranteed by the classification structure established in this report. Thus, the classification theorem not only solves the isomorphism problem in principle but also serves as the necessary foundation for further algebraic invariants in the theory.

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